Ch. 5 — P02 (A) Little's Law Applications

Problem Bank | Chapter 5 | Type: A | Difficulty:

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Apply Little's Law ($L = \lambda W$) to answer each question. Show your work.

Part A — Call center: A call center receives 120 calls per hour. The average call duration (time in system) is 4 minutes. How many agents must be busy on average?

Part B — Hospital: A hospital's emergency department sees 200 patients per day (24-hour operation). The average length of stay is 3 hours. What is the average census (number of patients in the ED)?

Part C — Manufacturing: A production line has an average work-in-progress (WIP) inventory of 80 jobs. The average throughput is 20 jobs per hour. What is the average cycle time (time from job start to completion)?

Part D — Inverse problem: A queueing theorist observes that $L = 15$ and $W = 3$ hours. She wants to reduce W to 2 hours without changing the arrival rate. What must L become? Is this achievable by reducing service time only? Explain.

Part E — Validity check: A student reports: "I simulated a checkout system and measured $L = 3.2$ customers, $\lambda = 10$/hour, and $W = 18$ minutes." Is this consistent with Little's Law? If not, identify the likely error.